Kudum Veerabhadraiah

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SUMMARY

Software Engineer, Data Scientist, and AI Engineer fresher with a B.Tech in Electronics & Communication Engineering and hands-on Data Science training. Strong foundation in software development, Python, SQL, machine learning, deep learning, artificial intelligence, data analytics, NLP, computer vision, APIs, cloud computing, and MLOps.

Experienced in designing and developing end-to-end technical solutions, from data collection, preprocessing, exploratory analysis, feature engineering, model development, evaluation, and visualization to API integration, cloud deployment, and production-oriented workflows.

Hands-on experience with Python, SQL, REST APIs, Flask, FastAPI, Git/GitHub, Docker, AWS, MLflow, PostgreSQL, MySQL, MongoDB, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Power BI, NLP, LLMs, RAG, embeddings, vector databases, and AI agents.

Built projects spanning crop-yield prediction, real-time obstacle detection, computer vision, rice leaf disease detection, handwritten digit classification, banking analytics, AI assistants, and AI-powered career platforms.

Seeking entry-level opportunities in Software Engineering, Data Science, Artificial Intelligence, Machine Learning, or Data Analytics where software engineering and AI capabilities can be combined to build reliable, scalable, and measurable solutions.

EDUCATION

Annamacharya Institute of Technology & Sciences, Andhra Pradesh Andhra Pradesh

Bachelor of Technology — Electronics & Communication Engineering, Engineering Mathematics, Programming Fundamentals, Electronics, Communication Systems, Digital Systems, Technical Problem Solving, Analytical Thinking, Engineering Design *Graduated: 2025*

EXPERIENCE

Internship Experience

Your previous profile references AI/ML internship-related work, but the exact company name, internship title, dates, responsibilities, and verified outcomes are not available in the information I can safely confirm here.

- For the final master resume, this section should be populated only with verified internship details: Company, Role, Location, Duration, Technology Stack, Key Responsibilities, Projects, Measurable Results.
- This avoids inventing employment history.

PROJECTS

Crop Yield Prediction System | Role: Data Scientist / ML Developer. Technologies: Python, Pandas, NumPy, Scikit-learn, XGBoost, PCA, AWS S3, Power BI.

- Developed a machine-learning solution for crop-yield prediction using a large agricultural dataset.
- Processed and analyzed 50,000+ agricultural records.
- Performed data cleaning and exploratory data analysis.
- Engineered relevant predictive features.
- Applied PCA for dimensionality reduction.
- Developed an XGBoost regression model.
- Evaluated model performance using RMSE and R^2.
- Achieved approximately $R^2 = 0.85$.
- Reduced RMSE by approximately 23% through feature/model improvements.

- Used AWS S3 for cloud-oriented data storage.
- Created Power BI dashboards for analytical interpretation.

Real-Time Obstacle Detection System | Role: ML Engineer / Software Developer. Technologies: Python, Random Forest, Flask, AWS Lambda, IoT Sensor Data.

- Developed a real-time obstacle detection system using sensor-based data and machine learning for smart mobility applications.
- Processed sensor observations.
- Prepared features for machine learning.
- Developed a Random Forest classification model.
- Achieved approximately 88% accuracy.
- Reduced false positives by approximately 20%.
- Built a Flask REST API for model integration.
- Explored AWS Lambda for cloud execution.
- Designed the system around real-time prediction requirements.

Rice Leaf Disease Detection | Role: AI / Deep Learning Project. Technologies: Python, TensorFlow, Keras, CNN, OpenCV.

- Developed a computer-vision-based deep learning workflow for detecting diseases from rice leaf images.
- Prepared image datasets.
- Performed image preprocessing and normalization.
- Applied computer vision techniques.
- Developed a CNN architecture.
- Trained the model using TensorFlow/Keras.
- Evaluated image classification performance.
- Designed the workflow for automated disease identification.

Handwritten Digit Classification | Role: Machine Learning / Deep Learning Project. Technologies: Python, TensorFlow, Keras, MNIST.

- Implemented a handwritten-digit classification pipeline using the MNIST dataset.
- Loaded and prepared handwritten image data.
- Normalized image pixel values.
- Split data into training and testing datasets.
- Built a neural-network-based classification solution.
- Trained and evaluated the model.
- Studied the complete image-classification lifecycle.

Portuguese Bank Marketing Analysis | Role: Data Science Project. Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn.

- Analyzed customer and marketing campaign data to identify factors affecting marketing outcomes.
- Cleaned customer and campaign datasets.
- Conducted exploratory data analysis.
- Investigated customer behavior patterns.
- Analyzed relationships between campaign variables.
- Visualized important business patterns.
- Prepared data for machine-learning experimentation.

Edge-AI Obstacle Detection for Smart Mobility | Explored an edge-oriented machine-learning architecture for real-time obstacle detection and intelligent mobility use cases. Technologies: Python, Machine Learning, Computer Vision, Sensor Data, Edge AI.

Maxi AI Assistant | Category: Generative AI / AI Agents. Conceptualized a multi-model AI assistant capable of conversational interaction and advanced agentic workflows. Technologies: Python, LLMs, OpenRouter, Playwright, AI Agents.

- Multi-model LLM integration
- OpenRouter-based model access
- Standard conversational mode
- Advanced agent mode

- Browser automation
- Playwright integration
- Tool-based execution
- Multi-step AI workflows

CareerFlux AI | Category: AI Product / Generative AI. Conceptualized an AI-driven career platform focused on resume optimization, career guidance, job matching, and skill development.

- ATS resume builder
- Resume scoring
- Job-fit analysis
- AI career coach
- Personalized career roadmap
- Job recommendations
- Skill-gap analysis
- Subscription architecture
- Admin analytics
- Technology concepts: AI / LLM, Python, FastAPI, PostgreSQL, Supabase, React / Next.js, Stripe, Razorpay, Cloud deployment

TECHNICAL SKILLS

Python
SQL
R
Object-Oriented Programming
Data Structures & Algorithms fundamentals
Functions and Modules
Exception Handling
File Handling
API Integration
REST API Development
JSON
Async Programming
Backend Development
Debugging
Virtual Environments
Package Management
Flask
FastAPI
Uvicorn
Pydantic
Pandas
NumPy
Scikit-learn
XGBoost
LightGBM
TensorFlow
Keras
OpenCV
Pillow
NLTK

Transformers
Sentence Transformers
ChromaDB
FAISS
BeautifulSoup
PyPDF2
python-docx
openpyxl
HTTPX
Machine Learning
Deep Learning
NLP
Computer Vision
LLMs
Generative AI
RAG
Embeddings
AI Agents
AWS S3
Amazon EC2
AWS Lambda
Docker
MLflow
CI/CD
MySQL
PostgreSQL
MongoDB
SQLite
Supabase
Neon
Aiven
Power BI
Tableau
Matplotlib
Seaborn
Plotly
Git
GitHub
Playwright
Browser automation
PyAutoGUI

CERTIFICATIONS

DataMites — Data Scientist Training
IBM SkillsBuild — Python for Data Science

CAREER FOCUS

Primary Roles: Software Engineer; Data Scientist; AI Engineer; Machine Learning Engineer. Additional Target Roles: Python Developer; Backend Developer; Data Analyst; AI/ML Engineer; Generative AI Engineer; NLP Engineer; Computer Vision Engineer; Junior Software Engineer; Junior Data Scientist; Junior Machine Learning Engineer.

PROFESSIONAL STRENGTHS

Analytical problem solving; Software development mindset; Machine-learning workflow design; Data-driven decision making; Debugging and troubleshooting; Technical research; AI application development; API development; Data visualization; Continuous learning; System design thinking; Project-based learning; Cross-domain technical understanding.

SOFT SKILLS

Communication; Team Collaboration; Problem Solving; Analytical Thinking; Adaptability; Time Management; Self-Learning; Technical Documentation; Attention to Detail.

ADDITIONAL LEARNING AREAS

Generative AI; AI Agents; Google Cloud; AWS; IBM AI technologies; edX learning programs.

CERTIFICATION RULE

For submitted job applications, list only certifications for which you have the completed certificate or verifiable credential.

ENGINEERING + DATA SCIENCE WORKFLOW

Problem Definition → Requirements Analysis → Data / System Design → Data Collection → Data Validation → Data Cleaning → EDA → Feature Engineering → Model / Algorithm Selection → Implementation → Testing & Evaluation → API / Application Integration → Containerization → Cloud Deployment → Monitoring / Improvement

ATS KEYWORDS

Software Engineer; Software Developer; Python Developer; Backend Developer; Data Scientist; Machine Learning Engineer; AI Engineer; Artificial Intelligence; Machine Learning; Deep Learning; Generative AI; LLM; RAG; AI Agents; Python; SQL; R; Pandas; NumPy; Scikit-learn; XGBoost; LightGBM; Random Forest; SVM; Regression; Classification; Clustering; PCA; Statistics; Data Analysis; EDA; Feature Engineering; Feature Selection; TensorFlow; Keras; CNN; Computer Vision; NLP; NLTK; TF-IDF; Word2Vec; Transformers; Embeddings; FAISS; ChromaDB; FastAPI; Flask; REST API; PostgreSQL; MySQL; MongoDB; SQLite; Supabase; Docker; MLflow; AWS; S3; EC2; Lambda; Git; GitHub; CI/CD; Power BI; Tableau; Matplotlib; Seaborn; Plotly; Playwright.

MASTER RESUME USAGE

This document is the comprehensive source resume. For individual applications, create targeted versions for Software Engineer, Data Scientist, AI Engineer, and Machine Learning Engineer roles.

FINAL RESUME PRINCIPLE

Use this master resume as the complete technical inventory. For each application, keep only the most relevant skills, projects, certifications, and achievements for the job description. Never claim an internship, certification, technology, metric, or professional responsibility that cannot be demonstrated or verified.